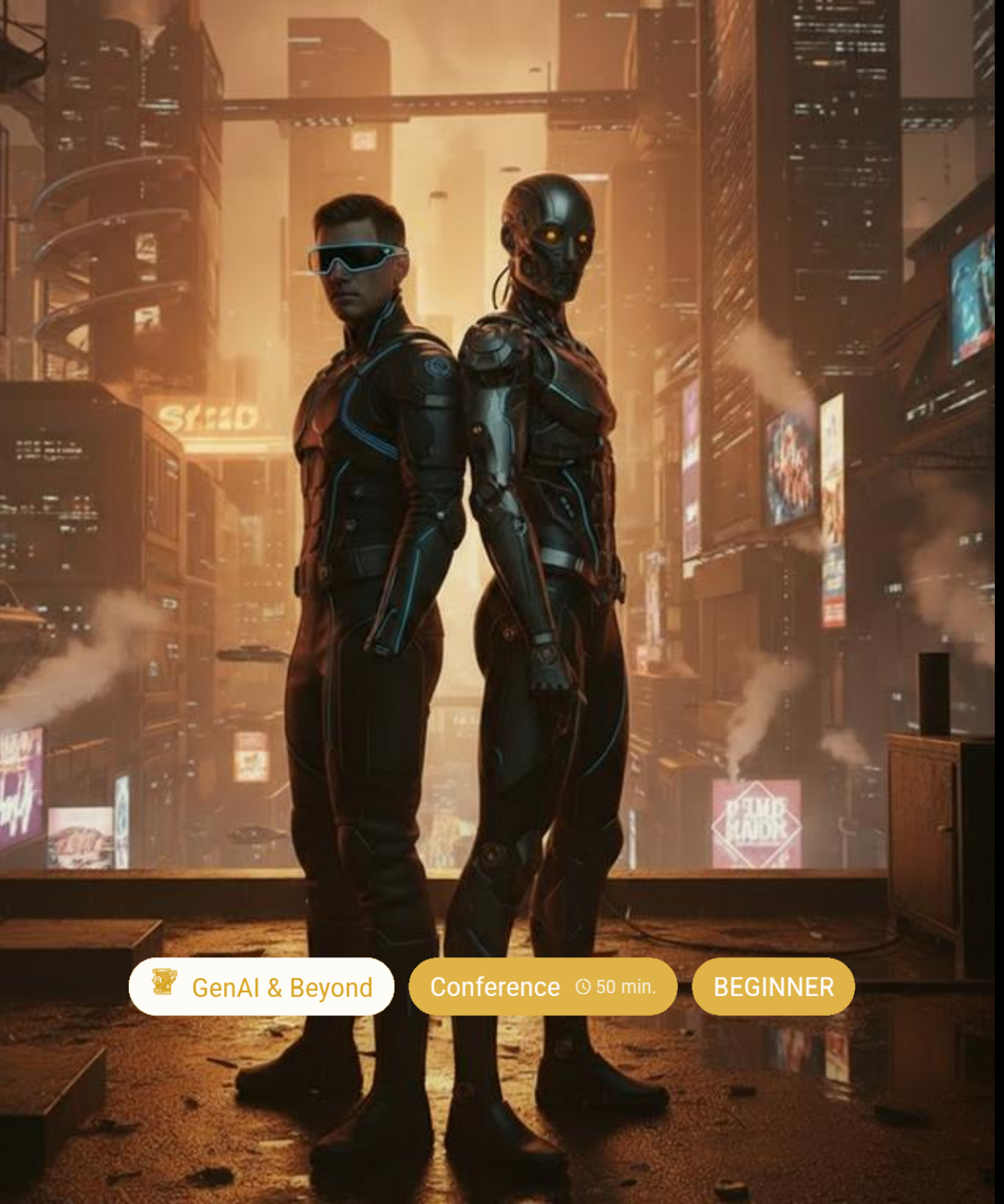


DEVOXX™

From OpenAI to OpenSource AI

Navigating between Commercial Ownership and Collaborative Openness

Raphaël Semeteys

2025 October 8

Raphaël Semeteys

Head of DevRel
Senior Architect, **Opensource Expert**

- Worldline since 1999
- Based in Paris
- raphiki.github.io



Contact

@RaphaelSemeteys

dev.to/raphiki/

linkedin.com/in/raphaelsemeteys/

PROJETS LIBRES PODCAST

Who are we?

We design payments technology that powers the growth of millions of businesses around the world.



7000+ engineers
in over 40 countries



Managing 33+ billion
transactions per year



€250M spent in products
and technology per year



Handling 150+
payment methods

DEVOXX™

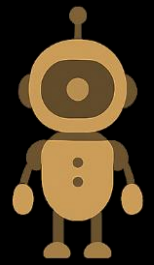
WORLDLINE

From simple statistical models to LLMs



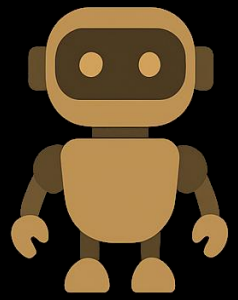
2010's

Word embeddings, Word2Vec, GloVe



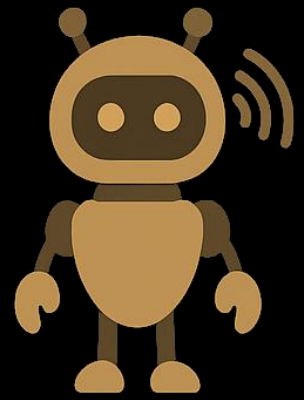
2017-2018

"Attention is All You Need", Transformers



2020's

ChatGPT, Multimodal, Responsibility



Tomorrow?

Small Language Models, Agents, Mobile & LAMs

GenAI is having its Linux Moment







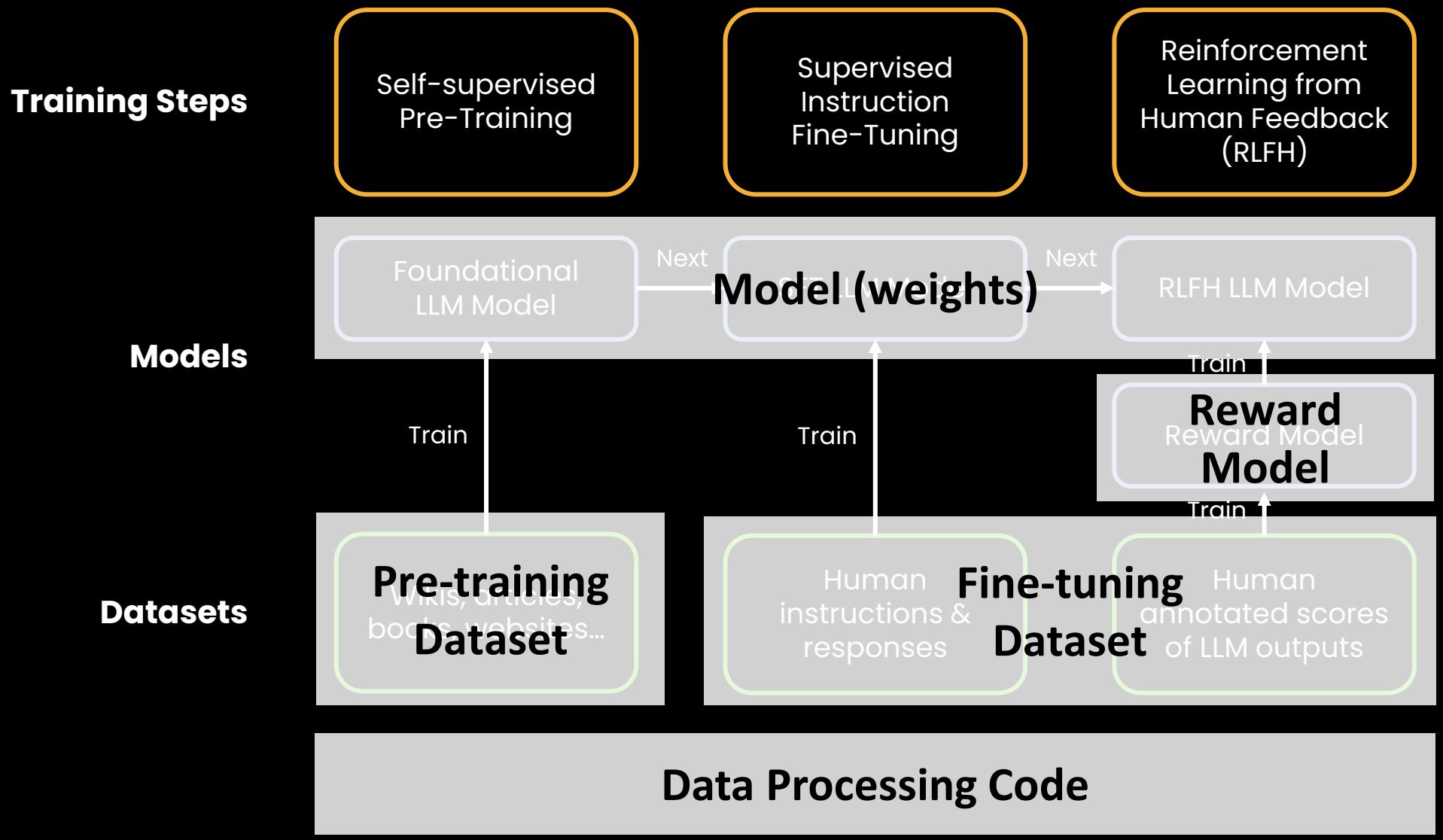


Just like open source (and the Internet), but on steroids!

Dynamics between Collaborative Openness and Commercial Ownership

Need for clarity on licenses

Openness of a Model



Openness of a Model

Score	Level	Description	Model (weights)	Pre-training Dataset	Fine-tuning Dataset	Reward Model	Data Processing Code
0	Closed	**No access** to public information, data or assets					
1	Published research	**Research papers** but without more information, data, or assets					
2	Restricted	**Access only with special agreement** (commercial, research, etc.)					
3	Open with limitations	**Access and reuse but with certain limitations of use**	Open Weights	Open Data			
4	Totally open	**Access and reuse without restrictions on use** (e.g. opensource license)					Open Source

Open Science

Big Player: Google

From open research to a closed and then pragmatic approach

Google BERT (2018-2019)

PaLM 2 (2022-2023)

Gemini (2024-2025)

Gemma (2024-2025)

Entreprise (US)		
Component	Score	Level
Model	4	Totally open
Dataset	3	Open with limitations
Code	4	Totally open

→

0	Closed

→

Open Weights

3	Open with limitations
1	Published research only
4	Totally open

Big Player: Google

From open research to a closed and then pragmatic approach

PaLM 2
(2022-2023)

Gemma
(2024-2025)

*You may **not use nor allow others to use** Gemma or Model Derivatives to: [illegals activities, unlicensed practices of profession, abuse, security bypass and promotion of hatred, abuse, violence, monitoring people without consent, misinformation/defamation, automate decisions concerning human rights and well-being, etc.]*

Component		
Model	3	Open with limitations
Dataset	1	Published research only
Code	4	Totally open

Responsible AI contradicts the Open Source Definition

Big Player: OpenAI

Deviation from the original vision of transparency and research

GPT 1 & 2
(2018-2019)

GPT-3.x, 4.x/o, Ox, 5.x
(2020-2025)

OpenAI

Generative
Pre-trained
Transforme
r

Non-Profit (US)		
Component	**Score**	**Level**
Model	**4**	**Totally open**
Dataset	**1**	**Published research only**
Code	**4**	**Totally open**

→

0	**Closed**

ChatGPT and o1
research articles only

Big Player: OpenAI

Deviation from the original vision of transparency and research

OpenAI

Generative
Pre-trained
Transformer

	GPT 1 & 2 (2018-2019)	GPT-3.x, 4.x/o, Ox, 5.x (2020-2025)
Component		
Model		0 — Closed
Dataset	1	Published research only
Code	4	Totally open

*You may not: […] Use **Output** to develop models that compete with OpenAI.*

ChatGPT and o1
research articles only



Prohibition of training commercial LLMs

Big Player: OpenAI

Open-weight reasoning models not served through the OpenAI API and are not available in ChatGPT

OpenAI

Generative
Pre-trained
Transformer

GPT-OSS

(June 2025)

Component	Score	Level
Model	4	Totally open
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	1	Published research only
Reward Model	1	Published research only
Code	4	Totally open

Open Weights

Big Player: OpenAI

Open-weight reasoning models not served through the OpenAI API and are not available in ChatGPT

GPT-OSS

(June 2025)

OpenAI

Generative
Pre-trained
Transforme
r

Apache 2 License

Usage Policy

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Reward Model	1	Published research only
Code	4	Totally open



Why did OpenAI waited 6 years to open again?

Big Player: Alibaba

Different path but same final positioning as other Big Players

Qwen & Qwen 1/2/3-72B

(2023-2025)

Enterprise (CN)		
Component	**Score**	**Level**
Model	**3**	**Open with limitations**
Pre-training Dataset	**1**	**Published research only**
Fine-tuning Dataset	**1**	**Published research only**
Reward Model	**1**	**Published research only**
Code	**4**	**Totally open**

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*You can not use the Materials or any **output** therefrom to improve any other large language model (excluding Tongyi Qianwen or derivative works thereof).*

Big Player: Alibaba

Different path but same final positioning as other Big Players

Qwen & Qwen 1/2/3-72B
(2023-2025)

Enterprise (CN)		
Component	**Score**	**Level**
Model	3	Open with limitations
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	1	Published research only
Reward Model	1	Published research only
Code	4	Totally open

Old or Large Models

+

Qwen 2, 2.5 & 3
(2024-2025)



4	Totally open
1	Published research only
1	Published research only
1	Published research only
4	Totally open

Medium Models

+

Qwen 3-max, Plus APIs
(2025)

0	Closed

Very Large Models

Other Big Players

Catching up in the GenAI **Gold Rush**

Create their own Models

GPT
(closed)



Microsoft Azure

Phi
(permissive open-weight)

Claude
(closed)





Titan/Nova
(closed)



Granite
(permissive open-weight)



OpenELM
(full open foundational)

Grok
(restricted open-weight)



ORACLE

Partner for Infrastructure (inference and training)

Challenger: Meta

Journey to openness

RoBERTa
(2018-2019)

Entreprise (US)		
Component	Score	Level
Model	4	Totally open
Dataset	1	Published research only
Code	4	Totally open

→

LLaMA by Meta
(2018-2025)

Component	Score	Level
Model	3	Open with limitations
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	1	Published research only
Reward Model	1	Published research only
Code	4	Totally open



Meta has always released open ~~source~~ weights models

Challenger: Meta

Journey to openness

LLaMA by Meta (2018-2025)

RoBERTa

Compon		
Model		
Dataset		
Code	4	Totally open

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		with tions
		shed h only
		shed h only
		shed h only
Code	4	Totally open

Use restrictions

License for platforms with +700 M users

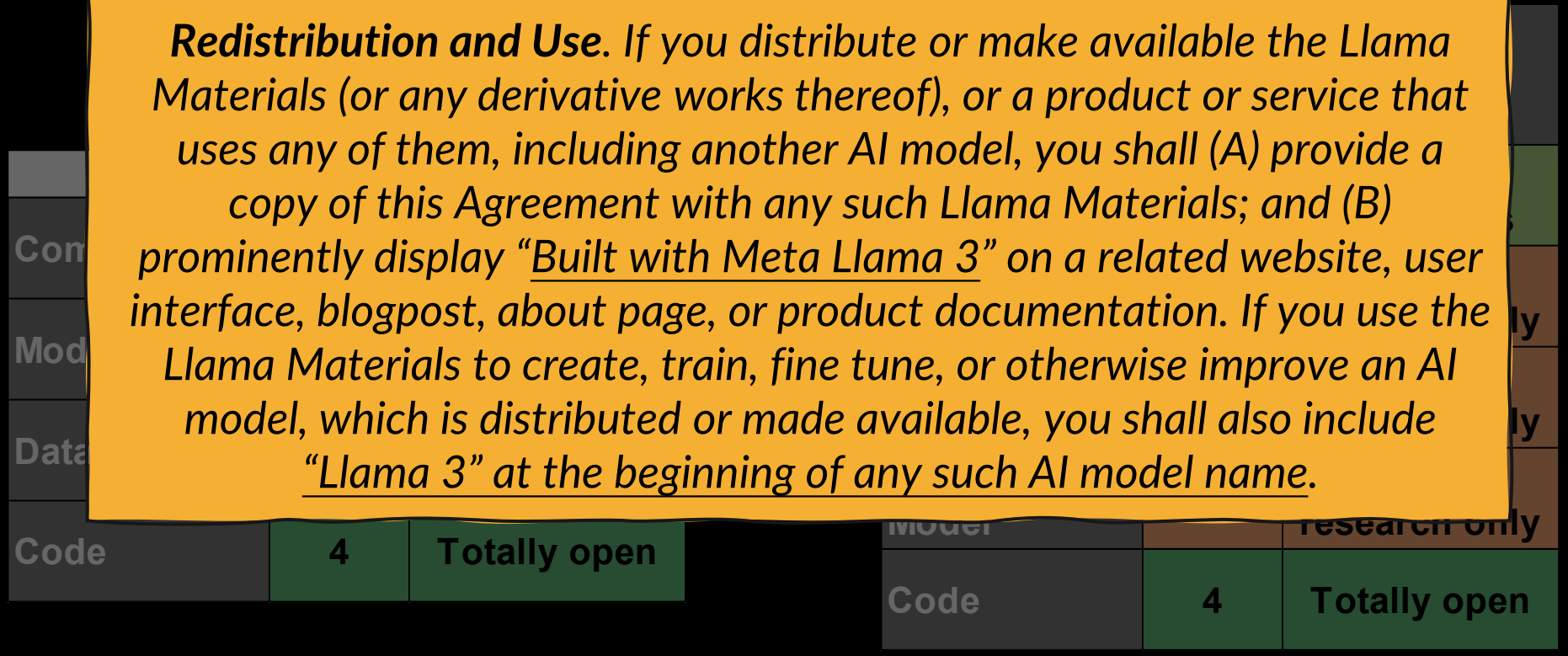
Challenger: Meta

Journey to openness

LLaMA by Meta (2018-2025)

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Code	4	Totally open



LLaMA 3 is **more restrictive** regarding redistribution and reuse

Challenger: Meta

Journey to openness

LLaMA by Meta (2018-2025)

RoBERTa

Compo		
Model		
Dataset		
Code	4	Totally open

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		with ons
		ed only
		ed only
Model		ed research only
Code	4	Totally open



LLaMA 4 is **banned for use** in Europe

Offspring of LLaMA: Alpaca and Vicuna

Research-only LLaMA fine-tuning by universities

Stanford Alpaca



VICUNA

Universities (US)		
Component	Score	Level
Model	2	Restricted
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	3	Open with limitations
Code	4	Totally open

Restrictions **inherited** from LLaMA 2 and OpenAI (ShareGPT)

Offspring of LLaMA: Deep Gogito

More recently: fine-tuning by a company

Deep Cogito v1
(April 2025)

Enterprise (US)		
Component	**Score**	**Level**
Model	3	Open with limitations
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	1	Published research only
Reward Model	1	Published research only
Code	4	Totally open

→

Deep Cogito v2
(July 2025)

Model	4	Totally open

Changed foundational model to Qwen



DEEP COGITO



Initial restrictions **inherited** from LLaMA 3.2
Proprietary IDA process **ported** to Qwen

Other Open Weights Challengers





Enterprises (CN/FR)		
DeepSeek / Mistral		
Model	4	Totally open
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	1	Published research only
Code	4	Totally open

DeepSeek

- Permissive Open Weight reasoning model
- Distilled models retain their original licenses: Qwen 2.5, LLaMA 3.1/3.3

Mistral

- Dual Strategy: Open Weights / APIs (*large*)
- Sustainable openness (*codestral*)

Other Open Weights Challengers





E		
De		
Model		
Pre-training Dataset		
Fine-tuning Dataset		
Code	4	Totally open

MNPL - 3.2. Usage Limitation

- You shall only use the Mistral Models and Derivatives (whether or not created by Mistral AI) for testing, research, Personal, or evaluation purposes in Non-Production Environments;

- Subject to the foregoing, You shall not supply the Mistral Models or Derivatives in the course of a commercial activity, whether in return for payment or free of charge, in any medium or form, including but not limited to through a hosted or managed service (e.g. SaaS, cloud instances, etc.), or behind a software layer.

ning model

iginal

/3.3

- Dual Strategy: Open Weights / APIs (*large*)
- Sustainable openness (*codestral*)



Rise of mixed **business models** (as in open source)

Open Foundational LLMs

	Research (UAE)		Research (EU)		Non-Profit (US)		Research (US)		Consortium (FR)	
	Falcon		BLOOM		EleutherAI		OpenLLaMA		Lucie	
Model	3	Modified Apache 2 License	3	OpenRAIL license	4	Apache 2 license	4	Apache 2 license	4	Apache 2 license
Pre-training Dataset	3	RefinedWeb methodology	4	ROOTS Corpus	4	The Pile	4	RedPajama	4	Lucie Training Dataset
Code	4	Totally open	4	Totally open	4	Totally open	4	Totally open	4	Totally open



Dataset fuzziness: refer to the specific license of the subset used

Responsible AI

Open Foundational LLMs

	Research (UAE)	Research (EU)	Non-Profit (US)	Research (US)	Consortium (FR)	
	Falcon	BLOOM	EleutherAI	OpenLLaMA	Lucie	
Model	3				4	Apache 2 license
Pre-training Dataset	3				4	Lucie Training Dataset
Code	4				4	Totally open

Falcon License

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Modified Open Source licenses

Open Fine-Tuned LLMs

	Enterprise (US)		Enterprise (US)		Enterprise (US)		Consortium (UAE/US)		Research (US)	
	Dolly		BLOOMChat		Zephyr		LLM360		OLMo-Instruct	
Model	4	Based on Pythia	3	Based on BLOOM	4	Based on Mistral	4	Totally open (Apache 2)	4	Totally open (Apache 2)
Pre-training Dataset	4	Based on Pythia	4	Based on BLOOM	1	Based on Mistral	4	Chunked and tokenized	4	Dolma (now ODC-BY)
Fine-tuning Dataset	4	Totally open CC BY-SA 3.0 License	3	Open with limitations	3	Open with limitations (OpenAI Data)	4	Totally open	4	Tulu (now ODC-BY)
Reward Model	N/A	No RLHF	N/A	No RLHF	1	Published research only	N/A	No RLHF	4	UltraFeedback (MIT)
Code	4	Totally open	4	Totally open	4	Totally open	4	Totally open	4	Totally open (Apache 2)

Impact of fundamental models and datasets

Open Fine-Tuned LLMs

	Enterprise (US)	Enterprise (US)	Enterprise (US)	Consortium (UAE/US)	Research (US)
	Dolly	BLOOMChat	Zephyr	LLM360	OLMo-Instruct
Model	4	totally open	4	totally open	4	totally open	4	totally open	4	totally open (Apache 2)
Pre-training Dataset	4	totally open	4	totally open	4	totally open	4	totally open	4	Dolma (now ODC-BY)
Fine-tuning Dataset	4	totally open	4	totally open	4	totally open	4	totally open	4	Tulu (now ODC-BY)
Reward Model	4	totally open	4	totally open	4	totally open	4	totally open	4	raFeedback (MIT)
Code	4	totally open	4	totally open	4	totally open	4	totally open	4	totally open (Apache 2)

Old AI2 ImpACT Licenses - Restrictions

[…] a. military weapons purposes […]
b. purposes of military surveillance […]
c. purposes of generating or disseminating information or content […] without expressly and intelligibly disclaiming that the text is machine generated;
d. purposes of 'real time' remote biometric processing […]
e. fully automated decision-making without a human in the loop […] as spreading misinformation […]
f. purposes of the predictive administration of justice, law enforcement, immigration, or asylum processes, such as predicting an individual will commit fraud/crime



Responsible AI limitations sometimes removed

Open Fine-Tuned LLMs

	Dolly		BLOOMChat		Zephyr		LLM360		OLMo-Instruct	
	Enterprise (US)		Enterprise (US)		Enterprise (US)		Consortium (UAE/US)		Research (US)	
Model	4	Based on Pythia					Totally open (Apache 2)	4	Totally open (Apache 2)	
Pre-training Dataset	4	Based on Pythia					Totally open	4	Dolma (now ODC-BY)	
Fine-tuning Dataset	4	Totally open CC BY-SA 3.0 License					Totally open	4	Tulu (now ODC-BY)	
Reward Model	N/A	No RLHF	N				No RLHF	4	UltraFeedback (MIT)	
Code	4	Totally open	4	Totally open	4	Totally open	4	Totally open	4	Totally open (Apache 2)

To monitor: Open-R1, OpenSeek



OpenSeek



Openness and transparency foster **collective innovation**

Global Overview

Open-Weight Era

🔒 Closed API-Only

Black Box

Access exclusively through commercial API endpoints

Gemini, GPT 3-5, Qwen3-Max, Mistral Large

Flagships Remain Closed

⚠️ Restricted Open-Weight

Licensing Restrictions

Weights available but legal use is constrained

LLaMA, Qwen (MAU Cap), BLOOM (RAIL License)

License is Everything

✅ Permissive Open-Weight

Private Data

Commercial freedom, data origins are non-disclosed

Gemma, GPT-OSS, Qwen, Mistral, DeepSeek, Falcon

Shovel Sellers & Challengers

🥇 Truly Open

Full Reproducibility

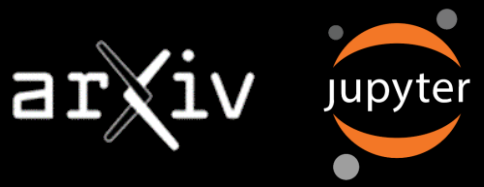
Scientific research, auditability, and trust

EleutherAI, OLMo, LLM360, Lucie

True Reproducibility is rare

Other aspects of the Linux Moment

Collaborative Tools and Ecosystems





Notebooks
Communities
New Business Models

Optimizations and Deployment





AI Chips, S/TLM
Distillation, Quantization
k8s, Decentralization

Open source Tools and Frameworks





Don't reinvent the wheel
Interop standards (OAI API, MCP)
Beyond Python

Initiatives to evaluate of LLM openness



open source initiative®

The Open Source AI Definition

THE LINUX FOUNDATION



MODEL OPENNESS FRAMEWORK



Center for Research on Foundation Models

The Foundation Model Transparency Index



GOUVERNEMENT

Liberté
Égalité
Fraternité

IA Model Openness Comparator

Let's Wrap-Up

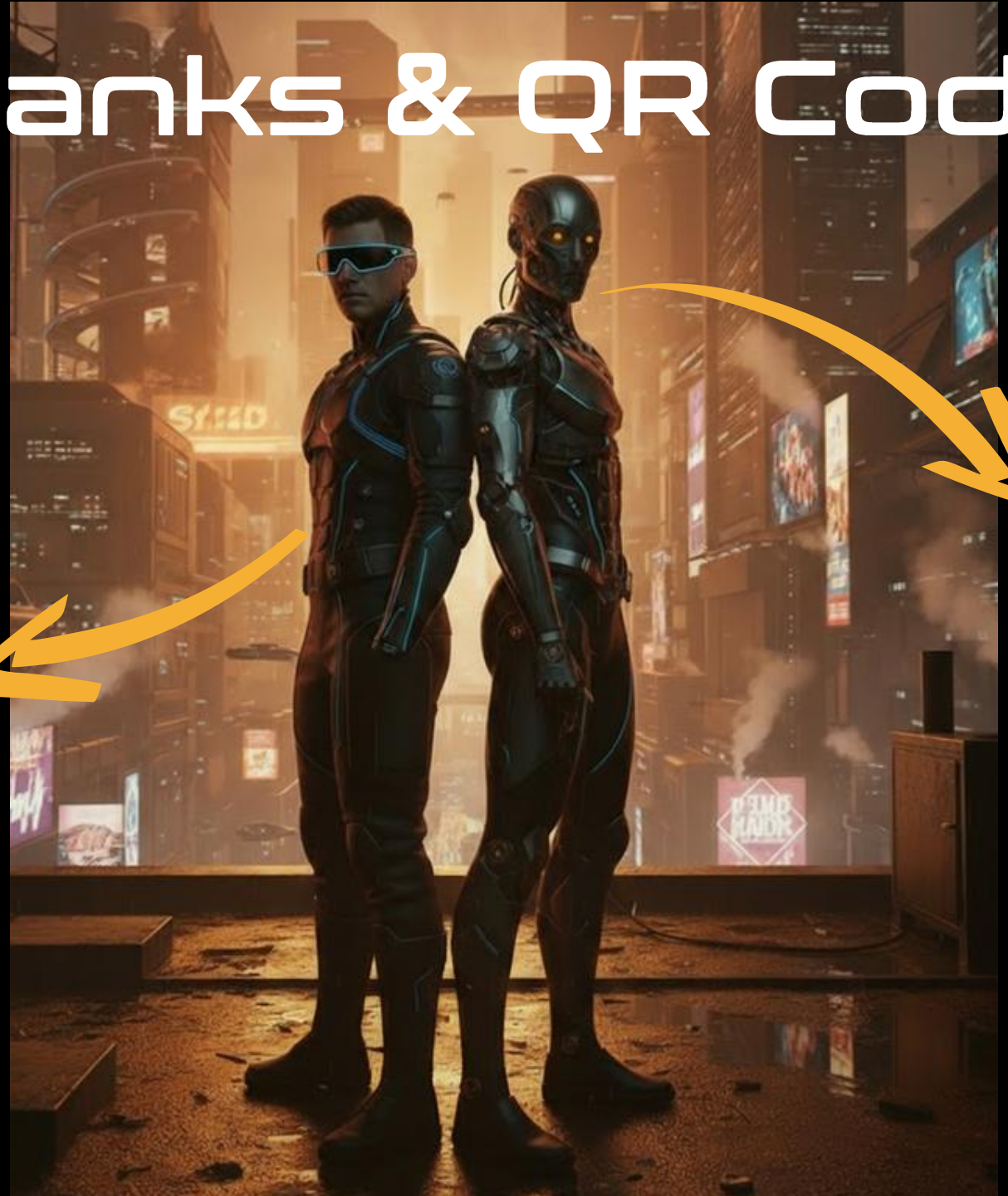
- **Open Research → APIs → Open Weights → Free AI** (as in *freedom*)
  - Datasets and upstream transitivity
  - Competition clauses
  - Restrictions related to Responsible AI
- **Competitive Market→ Cooperative Ecosystem**
  - Openness promotes reuse and collaboration
  - Collaboration leads to commoditization and more innovation



Linux Moment → **Linux Movement!**



Thanks & QR Codes



Your Feeback

LLM Openness **Agent**

DEVOXX

WORLDLINE